Pool-ball localisation and recognition

Computer Vision (CSU44053)
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Introduction

The aim was to locate pool balls in photographs and identify their type and number. I used `AllBalls.jpeg` to select detection parameters and build reference colour models, then repeatedly evaluated and refined the approach on Tables1–6.

1 Theory and application

1.1 Table masking and ball detection

The pipeline detects circles, extracts ball crops, estimates type and colour, assigns identities jointly, then transforms the results into an overhead view. Detection and colour analysis use the original photograph, since rectifying it first could distort circular ball outlines.

The supplied table corners, clockwise from top-left, are $(337, 35)$, $(1570, 21)$, $(1888, 451)$ and $(38, 472)$. I used their quadrilateral to mask the background, converted the image to greyscale and applied a 5×5 Gaussian filter with $\sigma = 1.5$ to reduce small intensity variations.

I used a gradient-based Circular Hough Transform to estimate centres and radii. After trying parameters manually, I searched 480 combinations, keeping radii within 15–30 pixels. Configurations were ranked by matched balls, then false positives, then mean centre error. Recall came first because a missed ball cannot be recognised later. The selected accumulator resolution ratio was 1.5, minimum centre separation 18 pixels, edge threshold 75 and circle-vote threshold 35.

Pockets can also produce circular detections. I transformed candidate centres into table coordinates and rejected those within 25 units of six approximate pocket centres. Each remaining ball was cropped from the original colour image and given a circular mask for recognition.

Table mask | Greyscale and smoothing

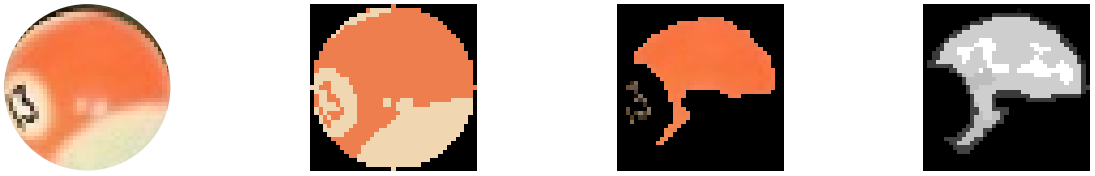
Hough circles | Pocket filtering

Figure 1: Detection stages on AllBalls. Pocket filtering removes the extra circle.

1.2 Ball type and body-colour histograms

I clustered the pixels inside each circular crop in BGR colour space using $K = 2$. The idea was that a striped ball's white and coloured regions would often form separate groups. I used three initialisations with a fixed random seed, stopping after 50 iterations or when cluster centres changed by less than 0.2. Each cluster's mean colour was converted to HSV: $V \geq 170$ and $S \leq 165$ indicated light pixels; a non-light cluster with $V \leq 90$ was dark. These thresholds were chosen manually. Cluster sizes then estimated the light fraction w.

For colour recognition, I built eight reference HSV histograms from labelled AllBalls examples, sharing a model between solid and striped partners and excluding the cue. A 3×3 elliptical erosion reduced boundary pixels, and bright, low-saturation pixels were removed to suppress white patches and reflections. Crops with fewer than 12 usable body pixels supplied no numbered-colour evidence.



Circular colour crop | Two colour clusters | Retained body pixels | Reference backprojection

Figure 2: Recognition stages for Table3 candidate 8. Backprojection visualises colour agreement; recognition uses direct histogram comparison.

Each histogram has $18 \times 8 \times 8 = 1152$ bins. Hue spans $[0, 180)$, and saturation and value span $[0, 256)$, so bins cover 10 hue units and 32 saturation or value levels. Finer hue grouping helps distinguish body colours, while broader saturation and value groups allow lighting variation. This is a design compromise, not a claim that these bin counts are optimal.

For nearly grey pixels (BGR maximum minus minimum below 15), I set hue to zero because it is unstable at low chroma. A $[1, 2, 1]/4$ filter smooths neighbouring bins, wrapping hue and repeating the other endpoints. Histograms sum to one, with solid and striped references weighted equally. Candidate histogram p is compared with each reference q using one minus Bhattacharyya distance [1]:

$$s(p, q) = 1 - \sqrt{1 - \sum_b \sqrt{p_b q_b}}.$$

This produces eight colour similarities rather than one fixed colour label.

1.3 Type compatibility and identity assignment

Hard thresholds can give different types to balls with nearly identical light coverage. I therefore used soft compatibility scores with expected light-fraction intervals $[0, 0.30]$ for solids, $[0.30, 0.85]$ for stripes and $[0.85, 1]$ for cue. For type t and observed light fraction w:

$$g_t(w) = \exp\left(-\frac{d(w, I_t)^2}{2(0.10)^2}\right).$$

Here d is zero inside interval I_t and the distance to its nearest endpoint outside it. Compatibility is one inside the interval and decreases smoothly outside; the width 0.10 controls the decrease. This is a heuristic score, not a probability. Numbered-identity scores multiply colour similarity by type compatibility. The cue score is $wg_{\text{cue}}(w)$.

Each ball identity can appear at most once. I arranged candidate-to-identity scores in a matrix and used Hungarian assignment [2] to maximise their total while enforcing uniqueness. Extra dummy columns allow several candidates to receive `Unknown`, mainly for surplus candidates or those without usable evidence. This is not a calibrated confidence-based rejection rule.

Sometimes a detected circle enclosed a white interior patch rather than the whole ball. I expanded the recognition radius of initially white candidates by one pixel and recomputed their scores, keeping the original detection centre and radius. If several candidates were initially white, I reserved cue for the largest original Hough radius. This uses size as a clue to distinguish a whole cue ball from a smaller white patch.

Genuine cue: original radius 24.15 pixels

Competing patch: original radius 17.55 pixels

Figure 3: Initially white candidates on Table1, shown at equal display size. The smaller circle captures an interior region of a striped ball.

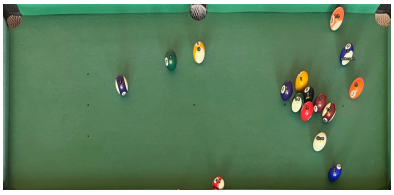
1.4 Coordinate transformation and visualisation

A homography maps the four table corners to $(0, 0)$, $(1433, 0)$, $(1433, 682)$ and $(0, 682)$ in a 1434×683 rectangle:

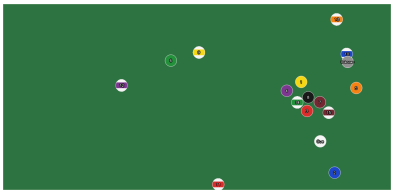
$$\lambda(x_t, y_t, 1)^T = H(x, y, 1)^T.$$

A table unit is approximately one millimetre at the supplied scale. Stored centres are transformed for matching and output. The overhead diagram uses fixed ball radii and colours/bands representing predicted identities, rather than observed orientation.

Original-image labels



Rectified labels



Overhead diagram

Figure 4: Transformation and presentation on Table2. All views retain the extra false detection.

1.5 Alternative approaches investigated

Fixed HLS thresholds. My first type estimator separated pixels into white, dark and coloured groups using HLS lightness and saturation. Their proportions supplied cue/solid/stripe rules. These were easy to interpret but sensitive to highlights and shadows, motivating K-means. The initial K-means rules labelled at least 85% light coverage as cue, at least 70% non-light coverage as solid, and other mixed cases as striped. I also tested $K = 3$–6.

Connected-component analysis (CCA). Colour proportions do not describe spatial arrangement. I therefore examined component counts, largest-region fractions and boundary light coverage. Larger white regions might distinguish stripes from solid balls' number patches, but highlights, orientation and thresholded shadows can change connectivity. Initial colour recognition used cluster-mean hue thresholds, with a separate low-value rule for black. Similar colours and shifting cluster means motivated the histogram descriptor.

HLS type predictions | Initial K=2 predictions | CCA light region

Figure 5: Alternative type evidence on Table1. A light region alone does not establish that a ball is striped or cue.

Segmentation-assisted detection. Touching balls interrupt each other's outlines. I tested foreground segmentation and distance-transform/watershed proposals near baseline detections, using area, shape, support, duplicate and pocket checks. A separate combined method estimated local cloth colour and routed foreground clusters to more permissive local Hough searches, retaining baseline circles outside these regions. Edge, arc and foreground checks filtered proposals; a subsequent marker-controlled watershed rejected circles with implausible supported basin areas. These checks used image evidence rather than ground-truth labels.

Foreground | Distance transform | Watershed basins

Combined method: local search regions | Combined method: watershed basins

Figure 6: Segmentation stages. Separating a region visually does not guarantee an accepted ball detection.

2 Results and analysis

2.1 Evaluation methodology

Tables1–6 contain 90 annotated balls. Predictions are matched one-to-one to annotated centres within 15 rectified table units. Matching prioritises the number of valid pairs, using proximity to resolve alternatives; it need not minimise total distance globally. Matched detections are true positives (TP), unmatched predictions false positives (FP), and unmatched annotations false negatives (FN). Pooled counts give:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

Mean localisation error is $\frac{1}{TP} \sum_i \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2$ for matched centres in table coordinates. Type accuracy measures cue/solid/striped correctness; identity accuracy requires the exact number or cue. If C identities are correct, matched identity accuracy is C/TP and end-to-end correct-identity recall is $C/90$. An Unknown label on a genuine ball is an identity error. Extra detections affect precision, but are not directly penalised by $C/90$.

2.2 Final performance on the supplied images

Table 1: Final detection and recognition results. Mean error is in table units.

Image	GT	TP	FP	FN	Mean error	Correct identities
Table1	16	14	1	2	7.68	11/14
Table2	16	16	1	0	3.58	16/16
Table3	16	15	0	1	3.12	15/15
Table4	15	14	0	1	3.89	14/14
Table5	14	14	0	0	5.83	13/14
Table6	13	13	0	0	5.76	11/13
Total	90	86	2	4	4.91	80/86

The detector achieves 97.73% precision, 95.56% recall and 96.63% F1. Identity accuracy is 93.02% on matched balls, while end-to-end correct-identity recall is 88.89%. The latter reflects the complete task because missed balls also count against it. All six genuine cues are identified correctly. The pooled mean centre error is 4.91 units, weighting images by their number of matches.

Table2 recognises all genuine balls but retains a false detection. Tables3 and 4 recognise every matched ball, yet each misses ball 10. Tables5 and 6 detect every ball but still make identity mistakes. Detection and recognition therefore remain separate sources of failure; high accuracy on matched balls does not imply that the whole scene is correct.

On AllBalls, the selected search configuration gave 16 TP, 1 FP and no FN (94.12% precision, 100% recall, 96.97% F1; mean error 2.76). Pocket filtering removed the false positive, and final recognition was 16/16. These are training-image results, since AllBalls supplied parameter selection and reference colours; they are excluded from the development totals.

2.3 What the experiments changed

Table 2: Detection alternatives on Tables1–6, using pooled counts.

Method	TP	FP	FN	F1 (%)
Selected global Hough + filtering	86	2	4	96.63
Watershed augmentation	86	2	4	96.63
Cluster-local Hough	86	16	4	89.58
Cluster-local Hough + watershed	83	11	7	90.22

Watershed augmentation accepted no detection changes. Local Hough retained the same total TP count but produced 14 more false positives than the selected detector. Its watershed check removed five false positives while losing three genuine matches. Although this improved F1 slightly over local Hough alone, both remained worse than the simpler global detector. A local recovery therefore did not establish an overall improvement.

Baseline: 15 TP, 0 FP, 1 FN Local Hough: 16 TP, 4 FP, 0 FN Watershed: 16 TP, 3 FP, 0 FN

Figure 7: Table3 illustrates a recovered ball accompanied by false detections.

Table 3: Recognition comparisons. Type and identity counts measure different tasks.

Task	Configuration	Correct /86
Type	HLS thresholds	43
Type	K=2	76
Type	K=3, 4 or 5 (each)	73
Type	K=6	74
Type	K=2 + CCA	73
Identity	Hue rules	68
Identity	HSV histograms, hard types	72
Identity	Independent soft scores	73
Identity	Add unique assignment	75
Identity	Add selective +1 crop expansion	79
Identity	Add largest-radius cue reservation	80

K=2 outperformed the tested larger cluster counts under the same type rules. Extra clusters can represent shading rather than useful material regions. CCA also reduced type correctness, so I retained the simpler estimator. The original controlled recognition comparisons held detections fixed: histograms added four correct identities, soft scoring one, assignment two, selective expansion four and cue reservation one. The total gain cannot be credited to histograms or assignment alone.

2.4 Successes and remaining failures

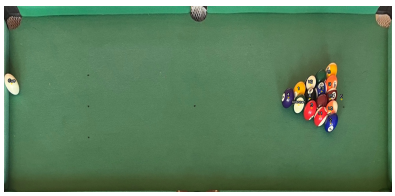


Table1: 11/14 identities; two misses

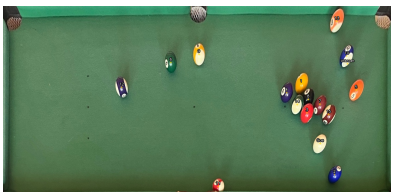


Table2: 16/16; one false detection

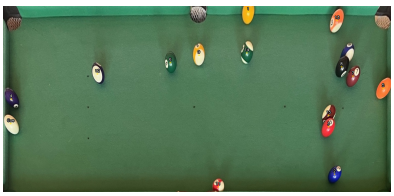


Table3: 15/15; one miss

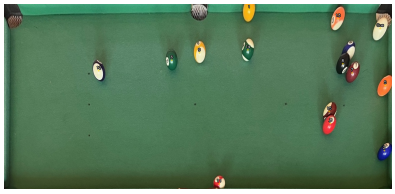


Table4: 14/14; one miss

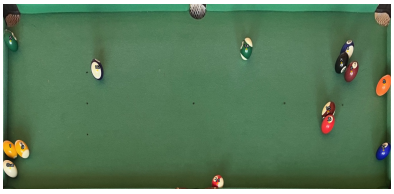


Table5: 13/14; no misses

Table6: 11/13; no misses

Figure 8: Final labelled results for all six development images. Fractions count correct identities among matched detections; labels include false detections and Unknown.

Crowding and localisation. Table1 misses balls 6 and 7 in the crowded group, assigns a false circle to 6, leaves ball 14 Unknown and swaps identities 5 and 9. Interrupted outlines make circle evidence weaker in crowded regions. The swapped matches have centre errors of 14.63 and 14.81 units, close to the 15-unit tolerance, so their recognition errors coincide with poor localisation. A displaced crop can include neighbouring colour or exclude part of a stripe, giving unreliable evidence even when a centre still counts as matched. Across all matches, median error is 3.75 units, maximum error is 14.81, and seven errors exceed 10 units.

Colour and type confusion. Tables5 and 6 label ball 10 as 4, changing both blue to purple and stripe to solid. Table6 also labels 3 as 11, preserving red but changing solid to stripe. The latter shows that correct body colour is not enough for correct identity. Stripe orientation can hide much of the white region, while reflections can make a solid appear lighter. These are plausible explanations for the confusions, rather than verified causes for each individual error.

Cue correction. Selective expansion improved the 75/86 control to 79/86 without losses, and cue reservation added the final correction. The row-adjusted radius rule made the same choices. Expanding every ball had worsened a separate baseline, so the benefit is specific to initially white crops. Assignment can also propagate a score change to other candidates.

Table1: crowding and false circle 6

Table6: ball 10 labelled 4

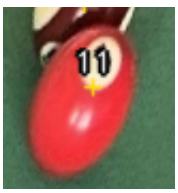


Table6: ball 3 labelled 11

Figure 9: Enlarged details of the final outputs make the remaining failures visible.

2.5 Robustness, appropriateness and improvements

The selected combination balances accuracy and complexity on the supplied images. Circular detection suits approximately round, visible balls, histogram distributions are less dependent on one representative colour, and uniqueness uses a real constraint of the task. The rejected alternatives did not improve their tested controls; this does not mean those methods are ineffective in other settings.

Limits of the evidence. Tables1–6 share a camera and broadly similar conditions, with some repeated arrangements. Repeated inspection and method selection introduce selection bias, even though histogram references came only from AllBalls. The reported scores describe development performance rather than performance on an untouched test set. One additional correct identity changes matched accuracy by about 1.16 percentage points, so small gains need further testing.

Changes in scene conditions. Fixed image-space radii, table corners and pocket locations depend on the camera setup. A new viewpoint or resolution would require updated geometry and radius ranges. Occlusion can remove the circular boundary needed by Hough detection. Different lighting, cloth colour or camera colour response can change histogram matches, and stripe orientation changes visible white coverage. The largest-radius cue rule can fail if a distant cue appears smaller than a nearby white patch. Evaluating new photographs with varied viewpoints, lighting and crowded arrangements, while freezing settings before inspection, is therefore the most useful next step.

Recognition improvements. More labelled reference examples under varied illumination and orientation could make the colour models less dependent on one training image. Better boundary estimation or cloth rejection could reduce crop contamination, but should be checked for both fixes and regressions. A calibrated Unknown policy could prevent weak candidates from taking real identities during assignment. The current best-effort assignment improves results, but its uniqueness constraint can also spread one candidate's error to another.

Evaluation improvements. The 15-unit tolerance is about 35% of the 43 mm ball diameter. Reporting performance across several tolerances would show how much success depends on accepting weakly localised matches; it should not be adjusted simply to raise the score. The homography describes the cloth plane, while visible ball centres lie above it, creating possible systematic error. Centre annotations also cannot establish the accuracy of masks or full ball boundaries. Boundary labels and isolated recognition comparisons would help distinguish detection geometry, colour evidence and assignment errors.

References [1] K. Dawson-Howe, CSU44053 course notes: *Images, Edges, Histograms, Region Segmentation, Learning and Evaluation*; based on *A Practical Introduction to Computer Vision with OpenCV*, Wiley (2014).

[2] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, 2, 83–97 (1955).